HEART DISEASE PREDICTION REPORT

Explainable Deep Learning Diagnosis Platform (CNN-LSTM-Attention + Grad-CAM)

Patient Index	#2	**True Label (Ground Truth)**	Normal
Model Prediction Probability	0.02%	**Analysis Date**	2026-06-23

Model Recommendation: NORMAL / LOW RISK

Patient Clinical Profile

Age	51 years	**Sex**	Male
Chest Pain Type (CP)	Non-anginal Pain	**Resting Blood Pressure**	130 mm Hg
Serum Cholesterol	256 mg/dl	**Fasting Blood Sugar (FBS)**	Normal (<=120 mg/dl)
Resting ECG	Left Ventricular Hypertrophy	**Max Heart Rate (Thalach)**	149 bpm
Exercise Induced Angina	No	**ST Depression (Oldpeak)**	0.5
ST Slope	Upsloping	**Vessels via Fluoroscopy (CA)**	0
Thalassemia (Thal)	Normal	**Predicted Diagnosis Probability**	**0.02%**

Primary Diagnostic Drivers (AI Focus)

1. CP

Patient Value: 3.0 | **Ideal Range:** 1, 2, or 3 (Angina/Non-anginal pain)

Interpretation: *Chest pain type 3 (Angina or localized chest pain).*

2. CA

Patient Value: 0.0 | **Ideal Range:** 0 vessels colored

Interpretation: *0 major vessels colored (clear arterial paths).*

3. SLOPE

Patient Value: 1.0 | **Ideal Range:** 1 (Upsloping)

Interpretation: *Upsloping ST segment (normal exercise response).*

4. RESTECG

Patient Value: 2.0 | **Ideal Range:** 0 (Normal)

Interpretation: *Abnormal ECG resting results (type 2), suggesting hypertrophy or ST-T wave anomalies.*

Secondary Clinical Abnormalities (Outside Top AI Focus)

1. TRESTBPS

Patient Value: 130.0 | **Ideal Range:** <120 mmHg

Interpretation: *Measured at 130.0 mmHg, indicating Stage 1 Hypertension.*

2. CHOL

Patient Value: 256.0 | **Ideal Range:** <200 mg/dl

Interpretation: *Measured at 256.0 mg/dl. High cholesterol level, indicating severe plaque build-up risk in coronary arteries.*

Explainability Visualization (Attention vs. Grad-CAM)

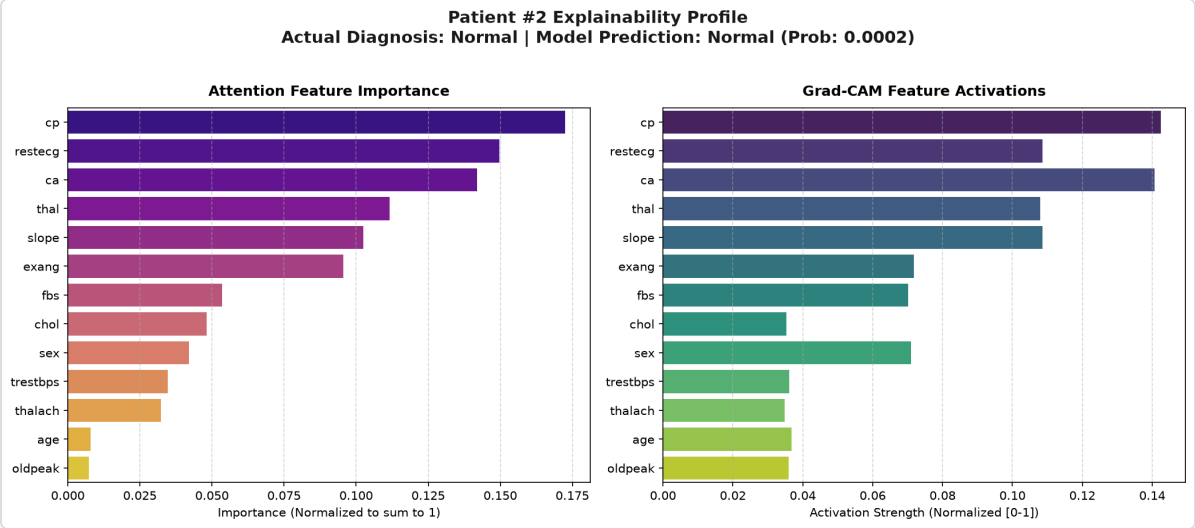


Figure 1: Dual-Perspective feature attribution showing global attention weight and patient-specific Grad-CAM contribution.

2D Gramian Angular Field (GASF) Heatmap

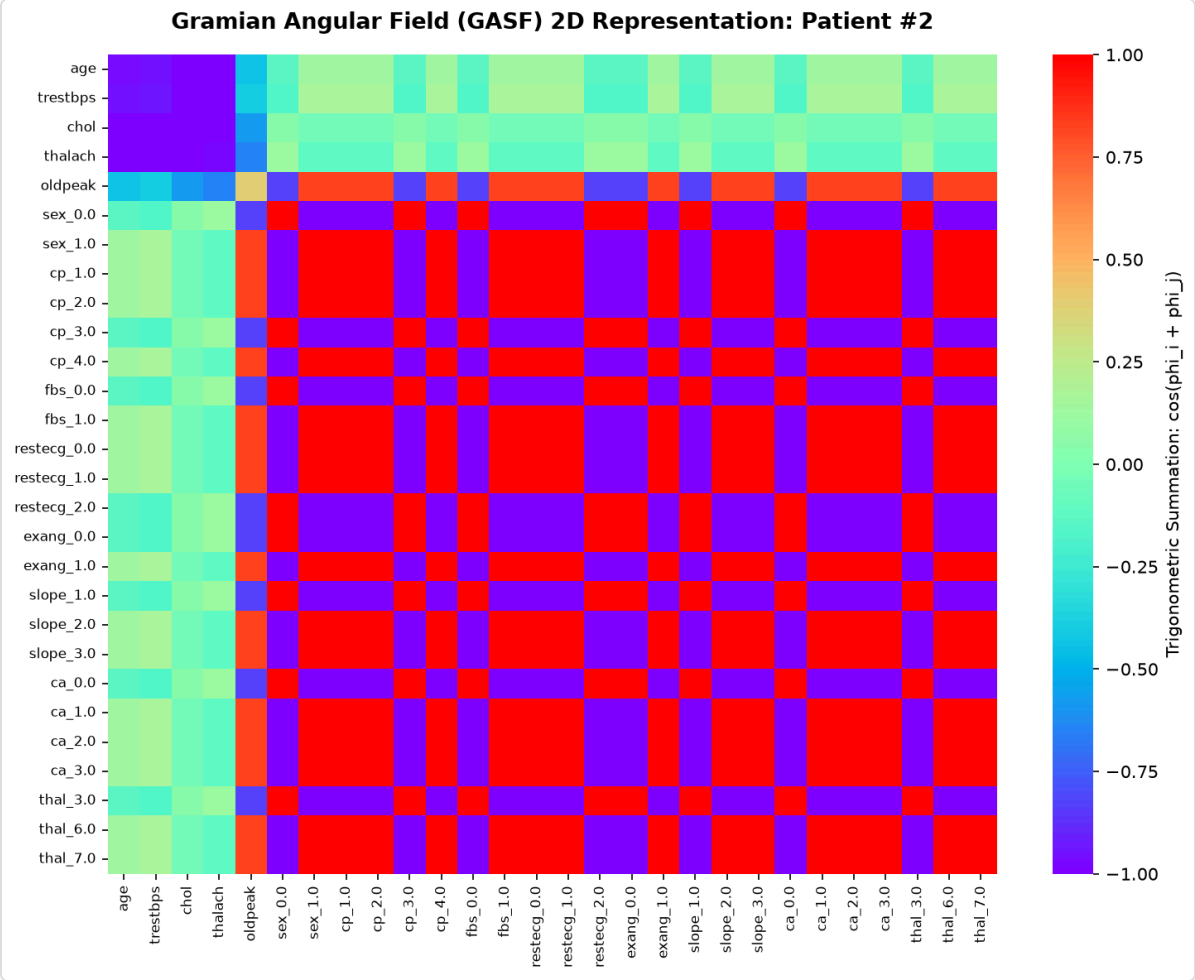


Figure 2: 2D feature-feature interaction texture mapping derived via Gramian Angular Summation Field conversion.